

15/16

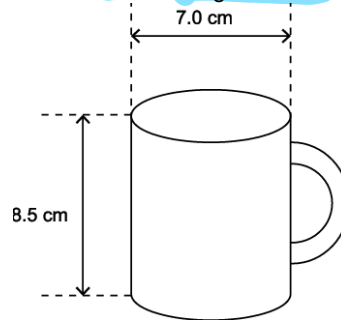
Cycle 2, Sept y13

10th lesson work – Specific Heat Capacity

Please see below the assigned work for the 10th lesson. This should take you approximately 1 hour and is designed to be completed in your physics study period – the allocated hour in your timetable with no teacher. You may choose to do this work digitally or on paper; in school or remotely.

The answers are included, for you to mark your own work. You are expected to show evidence of completion eg workings, as well as submitting your final score out of **16**. Please attempt all questions before going ahead to the answers (you will get more out of it if you do!).

- 1(a). A cylindrical cup of internal diameter 7.0 cm and height 8.5 cm is filled to the top with water.



The density of water is 1000 kg m^{-3} . The mass of one mole of water is 18 g. The specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$.

- Calculate the number of water molecules in the cup.

$$V = 8.5 \times 10^{-2} \times (3.5 \times 10^{-2})^2 \pi$$

$$\rho = \frac{m}{V} \Rightarrow m = \rho V = 1000V \text{ kg}$$

$$\text{num of moles} = n = \frac{m}{18 \times 10^{-3}}$$

$$\begin{aligned} \text{num of molecules} &= n \times N_A \\ &= n \times 6.02 \times 10^{23} \\ &= 1.09 \times 10^{25} \end{aligned}$$

$$\text{number of molecules} = 1.09 \times 10^{25} \quad [2]$$

- (b). Show that the minimum time taken for a 0.50 kW camping kettle to bring a cup of water at 20 °C to boiling point is about 200 s.

$$0.5 \text{ kW} = 0.5 \times 10^3 \text{ W} = 500 \text{ W}$$

$$\Delta \theta = 100 - 20 = 80$$

$$E = mc\Delta\theta \Rightarrow P = \frac{mc\Delta\theta}{t}$$

$$\Rightarrow t = \frac{mc\Delta\theta}{P} = 219.823 \dots \approx 200$$

[3]

- (c). In a laboratory test, the camping kettle was found to bring a cup of water to the boil in 320 seconds.

Explain why your previous answer is an underestimate and suggest **two** ways that you can refine the test to ensure that the time to boil is closer to 200 s.

The kettle isn't 100% efficient in heating the water as it also heats itself and the surrounding. The kettle could be pre-heated so less energy used heating itself. It could also be insulated so it doesn't heat up surrounding air.

[3]

- 2(a). A group of students conduct an experiment using water to heat glycerol in a boiling tube. The apparatus they use is shown in Fig. 20.1.

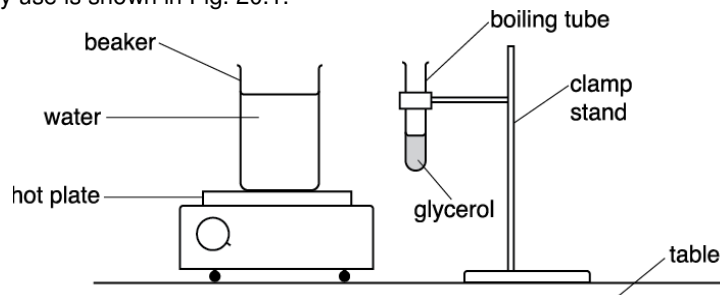


Fig. 20.1

The table below shows the mass m and the specific heat capacity c for water and glycerol used in the experiment.

	m / g	$c / \text{J kg}^{-1} \text{K}^{-1}$
Water	150	4200
Glycerol	20	2400

- i. The water is initially heated from 20 °C to 75 °C on a hot plate. Calculate the energy supplied to the water.

$$\Delta\theta = 75 - 20 = 55$$

$$E = mc\Delta\theta$$

$$= 150 \times 10^{-3} \times 4200 \times 55$$

$$= 34650 = 35 \times 10^3 \text{ J} \quad (2 \text{ sf})$$

energy = 35000 J [1]

- ii. The beaker of hot water at 75 °C is removed from the hot plate.
The boiling tube, which contains the glycerol at 20 °C, is now placed into the hot water.
Both liquids reach a common temperature θ .

Calculate the temperature θ .

$$34650 = (150 \times 10^{-3} \times 4200 + 20 \times 10^{-3} \times 2400) \Delta\theta$$

$$\Rightarrow \theta = 20 + \frac{34650}{(150 \times 10^{-3} \times 4200 + 20 \times 10^{-3} \times 2400)} = 71.106...$$

$$\theta = \dots\dots\dots 71^\circ \dots\dots\dots \text{°C [3]}$$

- iii. Explain why the actual temperature θ is different from your value calculated in (ii).

Some of the energy will be transferred to surroundings, not 100% efficient.

[1]

- (b). In a specialist laboratory, energy is supplied at a constant power to solid glycerol initially at a temperature of -100 °C. The glycerol is then heated from this temperature until it boils. The specific heat capacity of solid glycerol is less than the specific heat capacity of liquid glycerol. Glycerol melts at a temperature of about 20 °C and starts to boil at a temperature of about 290 °C.

Sketch a graph on Fig. 20.2 to show the variation of the temperature of glycerol with time. Assume that there is no heat transfer to the surroundings.

$5 \text{ kg}^{-1} \downarrow$

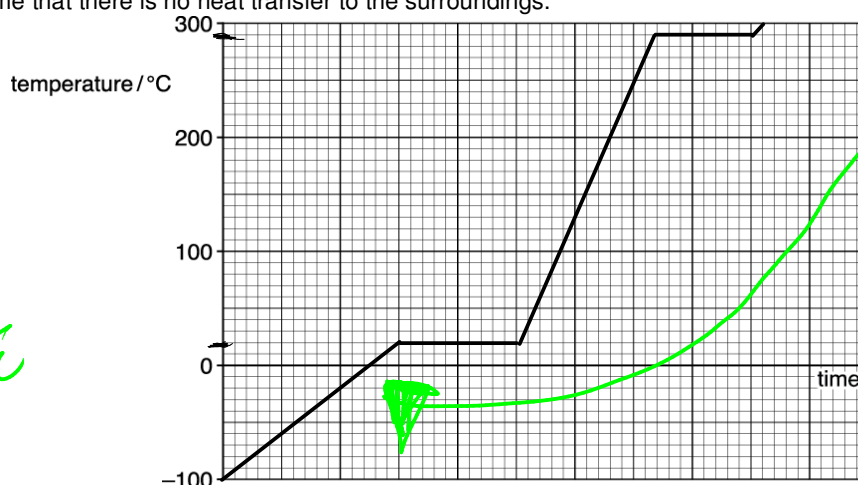


Fig. 20.2

$$E = mc\Delta\theta$$

$$\Rightarrow \Delta\theta = \frac{E}{mc}$$

$$\Rightarrow \Delta\theta \propto \frac{1}{c}$$

(If $c \downarrow$ then $\Delta\theta \uparrow$ and $m = \frac{\Delta\theta}{\Delta t} \Rightarrow m \uparrow$

This should be steeper, not other way around. This is because $\Delta\theta \propto \frac{1}{c}$

[3]

Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		number of moles = $0.327 / 0.018 = 18.17$	C1	
			number of molecules = $18.17 \times N_A$		
			number of molecules = 1.1×10^{25}	A1	
	b		energy input = $mc\Delta\theta = 0.327 \times 4200 \times 80 = 110 \text{ kJ}$	C1 M1	Allow 0.3 kg in the calculation
			energy input = power $\times$ time	C1	
			time = 220 (s)	A0	
	c		Thermal losses to kettle and surroundings	B1	My answer isn't here but seems sensible enough
			Lagging the kettle <i>(insulating)</i>	B1	
			Cover to prevent evaporation	B1	
			Total	8	
2	a	i	$E = m \times c \times \Delta\theta = 0.15 \times 4200 \times 55$ $E = 3.5 \times 10^4 \text{ (J)}$	A1	Note answer to 3 s.f. is $3.47 \times 10^4 \text{ (J)}$
		ii	(Energy transferred from water = energy transferred to glycerol) $0.150 \times 4200 \times (75 - \theta)$ or $0.020 \times 2400 \times (\theta - 20)$	C1	
		ii	$0.150 \times 4200 \times (75 - \theta) = 0.020 \times 2400 \times (\theta - 20)$	C1	
		ii	$\theta = 71(^{\circ}\text{C})$	A1	
		iii	The temperature is less / different because of thermal energy of the water is also used to warm up the boiling tube. (AW)	B1	
	b		Graph showing constant temperatures during phase changes. ✓	B1	
			Temperature increases linearly for the solid and the liquid. ✓	M1	
			Steeper slope for the solid state. ✗	A1	
			Total	8	